

Northwest Flight 85

Flight model: Boeing 747
Date: October 09, 2002

From: Detroit, US
To: Narita, Japan

There are 2 flight crew for this long flight.

Problem

The plane took a sudden roll the left and started to bank on the left. The pilot tries to level the plane, even though he is successful in doing so some thing is wrong. The computer shows that there is a problem with the rudder system. The long plane as this have 2 rudders which move together. But, in this case the lower rudder had shifted some degree to the left and gets stuck (**Hardover rudder**). The pilot tries to control the plane using the upper rudder and ailerons. Then the check the **COM (Cockpit operating manual)**, which contain list of procedures in case of any emergency situations. The final destination would take another 6 hours of flying time which is not possible with a jammed rudder. The plane tries to contact the last tower to perform an emergency landing, but the plane is situated in an dead zone in the middle of the sea and cannot contact any tower. The plane tries to contact the near by plane to relay distress the the nearest airport. The rudder is crucial for turning the plane. Without the rudder, the pilots had to control the plane direction using the ailerons which would lead the plane in the **slip**, which means the turn is not coordinated wherein the nose points to one direction and tail points to another direction. The nearest airport in close to 2 hours away. The captain declares the crew with an **red emergency** which means that the crew must start preparing for the worse. They then try to decent to a lower altitude so that the denser air could provide additional lift. They try to complete landing checklist way ahead of the near by airport so that they understand how their plane reacts in the case of landing. The limit their min altitude for the same to 14,000 feet which is considerably high enough to regain control if something goes wrong. The plane did respond positively, the pilots also tries to control the plane with the engine power. But, the problem is uneven engine power could increases the stress on the rudder which may damage is further evenly leaving the plane uncontrollable. Finally, the plane touches down the runway as planned earlier one of the pilots take control of the plane and other uses the **tiller** to control the nose wheel.

Investigation

The investigators find hydraulic fluid leaking from tail of the plane. When the tail section was opened it was found that an end cap of the power control module a part of the hydraulic unit that is used to controls the lower rudder was fractured. The cause of the fracture was metal fatigue. This very plane was the first of the model that was sold and was used as a test plane before it was sold. So their is a possibility either the plane should have been too old or put to vigorous test during the test flight causing higher stress leading to fracture. Later, an Air France flight had faced with the same problem. When examining that the investigators discover a manufacturing defect. They reported a sharp increase in radius at a particular cross section which could have been the initial point of fracture eventually propagating the fracture further. Also, the North west airline 85 did not crash dues to the split rudder in Boeing 747 which jammed only one rudder making the other functional.

Changes that were brought after the incident

The report advises the airlines to routinely inspect power control module. Also, they had recommended fail safe design of the power control unit after the air France incident. The board had also acknowledged the pilots decisions in the critical movements after the incident; that lead the plane to land safely and not plummet. Also, the importance of cockpit resource management was shown in the same flight. The pilots also stressed more on the importance of having good manual hand flying skills.

Reference

1. Rudder Failure at 35,000 Feet — Mayday Air Disaster
<https://www.youtube.com/watch?v=hK7E8AEQXrU&t=72s>